

Operators in string

1. Arithmetic:-

```
s1 = "Hello"
s2 = "World"

print(s1 + s2)
print(s1 * 3)

# output
HelloWorld
HelloHelloHello
```

2. Assignment:-

```
s1 = "Hello"
s2 = "World"

s1 += s2
print(s1)

s1 *= 2
print(s1)

# output
HelloWorld
HelloWorldHelloWorld
```

3. Comparison:-

```
s1 = "apple"
s2 = "banana"

print(s1 == s2)
print(s1 != s2)
print(s1 < s2)
print(s1 <= s2)
print(s1 > s2)
print(s1 >= s2)

# output
False
```

```
True
True
True
False
False
```

4. Identity:-

```
s1 = "hello"
s2 = "hello"

print(s1 is s2)
print(s1 is not s2)

# output:
True
False
```

5. Membership:-

```
s = "Python Programming"

print("Python" in s)
print("Java" not in s)

# output
True
True
```

6. Logical:-

```
# Non-empty string → True
# Empty string "" → False

# and
s1 = "Hello"
s2 = "World"
s3 = ""

print(s3 and s2)
print(s1 and s2)

# output
""
World
```

```
# or
print(s3 or s2)
print(s1 or s2)
#output
World
Hello
```

```
# not
print(not s1)
print(not s3)
# output
False
True
```